Chapter 2. Classical Option Pricing Theory

Financial Engineering

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Content

Probability

- Probability is the branch of mathematics concerning numerical descriptions of how likely an event is to occur, or how likely it is that a proposition (命题) is true.
- The higher the probability of an event, the more likely it is that the event will occur.
- Classical probability (古典概型) is a simple form of probability that has equal odds of something happening. It was founded in the 17th century, based on quantitative analysis in gambling.

Axiomatization

- Mathematicians spent nearly 300 years to define the probability concepts within a rigorously logic system.
- Enlightened by the developments of Lebesgue measure theory and functional analysis in the early 20th century, Soviet Mathematician A. Kolmogorov wrote the book *Foundations of the Theory of Probability*, in which the modern probability theory is constructed in an axiomatic system (公理化系统) like Euclid's *Elements* (几何原本).
- Modern probability theory with axiomatic system becomes a serious branch of mathematics, which developed fast in the 20th century.

Probability Space

- Kolmogorov's axiomatic Probability theory describes the uncertain information structure of the objective world.
- Three fundamental elements:
  - Sample space (样本空间) Ω
  - Class of events (事件族) $\mathscr{F}$
  - Probability measure (概率测度) P
- The triplet (三元组) $(\Omega, \mathscr{F}, P)$ forms a probability space (概率空间).

Sample space

- The sample space Ω is the starting point to consider probability problems, which contains all the possibility (randomness) of uncertain phenomenon.
- An element $\omega \in \Omega$ is called a sample (样本). Each sample corresponds to one possible randomness.
- The sample space is often quite abstract, due to its unobservability. Typically we specify a large enough sample space in each specific problem.

Finite Sample Space

- Example 1. Toss a coin. $\Omega = \{H, T\}$ or $\{1, 0\}$ or $\{$正, 反$\}$.
  - A structure. The names are not important.
- Example 2. Roll a die for N times. $|\Omega| = 6^N$.

$$\Omega = \{ \text{ All the strings of '1'–'6' with a length of } N \text{ } \}.$$

- It suffices to define the probabilities of every sample ω. Classical probability theory is strong enough in this case.

Infinite Sample Space

- Example 3. Select a natural number. $\Omega = \mathbb{N}$. The set Ω is countably infinite (可列无穷).
- Example 4. Toss a coin for infinite times.

$$\Omega = \{\text{ All the infinite string of 'H' and 'T' }\}$$

  which is uncountably infinite.
- Example 5. Select a value within $[0, 1]$. $\Omega = [0, 1]$.
- Classical probability theory fails in these scenarios, due to the lack of countable additivity (可列可加性)*.

σ-algebra

- Any subset $A \subset \Omega$ is known as an event (事件) in Ω.
- A class of events (子集族) forms a set $\mathcal{F}$. If this $\mathcal{F}$ satisfies
  - 1. $\phi \in \mathcal{F}$;
  - 2. "$A \in \mathcal{F}$" $\Rightarrow$ "$A^c \in \mathcal{F}$" ;
  - 3. "$A_1, A_2, \cdots, A_n, \cdots \in \mathcal{F}$" $\Rightarrow$ "$\bigcup_{n=1}^{+\infty} A_n \in \mathcal{F}$" ,

  then $\mathcal{F}$ is called a σ-algebra (σ-代数) or σ-field (σ-域) on Ω.
- "σ-" is only a kind of prefix, which indicates the closedness (封闭性) under countable union (可列并).

Examples of σ-algebra

- The minimum σ-algebra: trivial σ-algebra $\{\phi, \Omega\}$.
- The maximum σ-algebra: all the subsets
$$2^{\Omega} := \{A : A \subset \Omega\}.$$
- σ-algebra formed by segmentation: $A_1, A_2, \cdots, A_n$ are mutually exclusive (互不相交) subsets of Ω, with
$$\bigcup_{i=1}^{n} A_i = \Omega.$$
Then a σ-algebra can be formed by finite set operations (unions, intersections, ad complements) among $A_1, A_2, \cdots, A_n$. This σ-algebra contains exactly 2^n sets.

Measurable Space

- The doublet $(\Omega, \mathcal{F})$ is called a measurable space (可测空间). Any set $A \in \mathcal{F}$ is called a measurable set in $(\Omega, \mathcal{F})$, or an $\mathcal{F}$-measurable set.
- An $\mathcal{F}$-measurable set can become unmeasurable, if we replace $\mathcal{F}$) into a smaller σ-algebra.
- Measurable space prepares “a good structure” to be measured by measures.

Measure and Probability

- Given $(\Omega, \mathcal{F})$, a mapping (映射) $P : \mathcal{F} \to \mathbb{R}$ satisfying
  - 1. P always takes nonnegative values, and $P(\phi) = 0$;
  - 2. countable additivity: mutual exclusive subsets $A_1, A_2, \cdots$ have

$$P\left(\bigcup_{n=1}^{+\infty} A_n\right) = \sum_{n=1}^{+\infty} P(A_n),$$

  is defined as a measure (测度) on $(\Omega, \mathcal{F})$.

- Especially, if $P(\Omega) = 1$, the measure P is called a probability. The triplet $(\Omega, \mathcal{F}, P)$ is known as a probability space (概率空间).

Random Variable

- Given $(\Omega, \mathcal{F})$, such a mapping $X : \Omega \to \mathbb{R}$ satisfying

  $$\{\omega \in \Omega : X(\omega) \leq a\} \in \mathcal{F}, \qquad \text{for any } a \in \mathbb{R}$$

  is defined as a random variable (随机变量) or measurable mapping.

- We can write X instead of $X(\omega)$, if there is no ambiguity. But note that X is still a mapping.

- Proposition*. if X and Y are random variables, then $X+Y$, $X-Y$, XY, X/Y (if $Y \neq 0$), $\max(X, Y)$, and $\min(X, Y)$ are also random variables.

Distribution

- The distribution F_X of a random variable X maps any Borel* set $B \subset \mathbb{R}$ into

$$F_X(B) = P(X \in B).$$

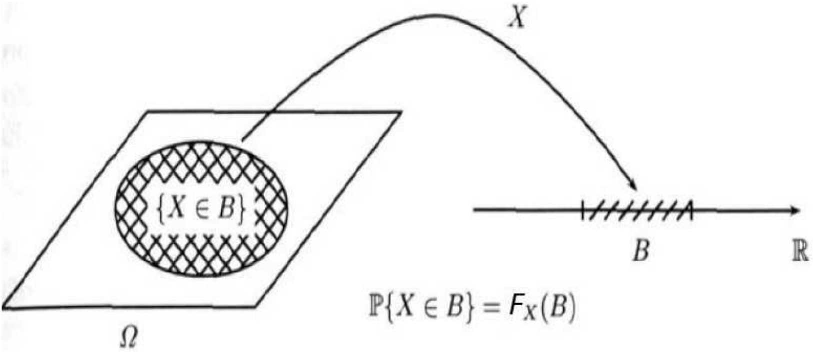
- Practically, it suffices to study all the “left half axis” $B = (-\infty, a\,]$ for $a \in \mathbb{R}$, since they can generate all the Borel set. Therefore, we define a cumulative distribution function (CDF, 累积分布函数) for X:

$$F_X(a) = P(X \le a),$$

which contains all the distribution properties of X.

Distribution

Distribution is a relation between $(\Omega, \mathcal{F})$ and $(\mathbb{R}, \mathcal{B})$.



Properties of CDF

- Monotonically increasing
- Right continuous, but not necessarily continuous
- Left limit 0; right limit 1
- Any random variable has a CDF.
- CDF of a discrete random variable is a step function.

Expectation

- The expectation (期望) of X is defined by the following Lebesgue* integral on Ω

$$E[X] := \int_{\Omega} X dP.$$

- Discrete random variable X:

$$E[X] = \sum_{k=1}^{+\infty} y_k P(X = y_k).$$

- Continuous random variable X:

$$E[g(X)] := \int_{\Omega} g(X) dP = \int_{-\infty}^{+\infty} g(x) f_X(x) dx,$$

where f_X is the probability density function of X.

Stochastic Process

- Given a measurable space $(\Omega, \mathcal{F})$, a stochastic process (随机过程) is a class of random variables $\{X_t : t \in \mathscr{T}\}$, where $\mathscr{T}$ is the set of time points.
- Typically we set $\mathscr{T} = \mathbb{N}$ or $[0, +\infty)$ in finance and other applied areas.
- A stochastic process is essentially a bivariate mapping from $\Omega \times \mathscr{T}$ to $\mathbb{R}$.
  - For fixed time $t \in \mathscr{T}$, $X_t(\omega)$ is a random variable on $(\Omega, \mathcal{F})$, denoted by $X_t \in \mathcal{F}$.
  - For fixed sample $\omega \in \Omega$, measurable function $X_\omega(t)$ is called a sample path (样本轨道).

Examples of Stochastic Processes

- Discrete time, $\mathscr{T} = \mathbb{N}$
  - Numbers of students in every lectures of this course
  - Seasonal GDP of China
- "Discrete time stochastic process" has the same definition with "time series", but the latter one emphasizes more on the statistical information of historical data.
- continuous time, $\mathscr{T} = [0, +\infty)$
  - Queue length in front an ATM
  - Numbers of broken lights in this classroom
  - Stock prices $\{S_t : t \geq 0\}$

Finite Dimension Distribution

- Consider a stochastic process $\{X_t : t \geq 0\}$. For any $n \in \mathbb{N}^+$, and any n time points $0 \leq t_1 < t_2 < \cdots < t_n$, the joint distribution of the n dimensional random vector

$$(X_{t_1}, X_{t_2}, \cdots, X_{t_n})$$

  is defined as a finite-dimensional distribution (FDD, 有限维分布) of $\{X_t\}$.

- All the FDDs of $\{X_t\}$ make up its class of FDDs (有限维分布族), partially capturing the property of $\{X_t\}$.

- Different stochastic processes can share the same class of FDDs.

Gaussian Process

- If all the FDDs of $\{X_t\}$ are all multivariate normal random vectors, then $\{X_t\}$ is known as a Gaussian process (高斯过程) or normal process.
- Quiz 4. Revision: what is multivariate normal distribution? How to express such distribution?
- Gaussian process includes Brownian motion. Such process can describe the noise in signal processing.
- For a Gaussian process $\{X_t\}$, two functions

$$m(t) := E[X_t], \quad K(s,t) := Cov(X_t, X_s),$$

can determine the entire class of FDDs.

History of Brownian Motion

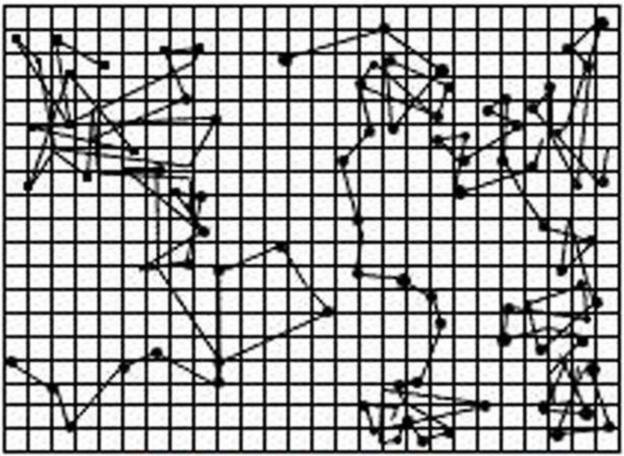
- In 1828, British Botanist R. Brown observed the random movement exhibited by plant pollens suspended in water.
- Brownian motion was noticed by physicians in late 19th century, which promoted the development of Kinetic Molecular Theory (分子运动论).
- In 1900, L. Bachelier introduced Brownian motion into Finance. This is regarded as the starting point of the field Financial Mathematics.

History of Brownian Motion (Cont'd)

- In 1905, A. Einstein proposed the first quantitative model of Brownian motion in terms of a Gaussian process, which explains some principles of molecular movements.
- In 1908, J. Perrin designed experiments that verifies Einstein's model, which won the Nobel Prize in 1926.
- In 1923, N. Wiener developed rigorous theoretical system of Brownian motion and relevant functional structures. Wiener's construction is basically equivalent with Einstein's definition, and is quite close to our modern definition.

Particle Movements

Three paths of particles observed by physicians



Brownian Motion

- A stochastic process $\{W_t : t \geq 0\}$ is called Brownian motion (布朗运动) or Wiener process if satisfying
  - $W_0 = 0$, continuous (连续) paths;
  - independent increments (独立增量):
    for $0 = t_0 < t_1 < \cdots < t_m$, the increments
    $W_{t_1} - W_{t_0}, W_{t_2} - W_{t_1}, \ldots, W_{t_m} - W_{t_{m-1}}$ are independent;
  - stationary increments (平稳增量):
    for $0 \leq s < t$, we have $W(t) - W(s) \sim \mathcal{N}(0, t - s)$.

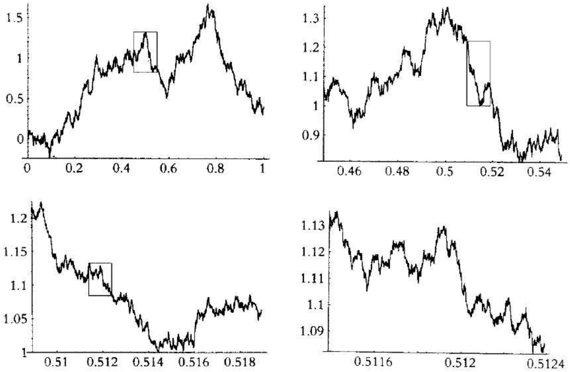
- Quiz 5. What is the main difference between this definition and that we learned in middle school physics?

Properties of Brownian Motion

- Markovian, which is weaker than independent increments.
- Gaussian: $(W_{t_1}, W_{t_2}, \cdots, W_{t_n})$ is multivariate normal.
- Quiz 6. Find $E[W_t W_s]$.
- Quiz 7. Find the distribution of $W_t + W_s$.
- Quiz 8. Compute $E[e^{\alpha t + \sigma W_t}]$.

Self-Similarity

An H-self similar process $\{X_t\}$ guarantees that for any $c > 0$, two processes $\{X_{ct}\}$ and $\{c^H X_t\}$ have the same class of FDDs.



Brownian Motion and Random Walk

- Let $\{X_n\}$ be a sequence of independent identical distributed (i.i.d.) random variables, and each term X_n takes values 1 or -1 with equal probabilities. Define $S_n := \sum_{k=1}^{n} X_k$, a simple symmetric random walk.
- Compressing. Define

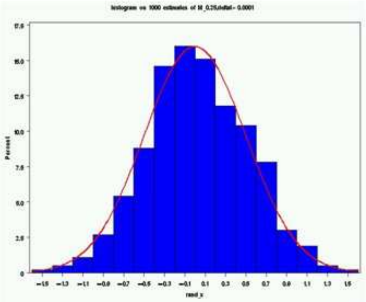
$$W^{(n)}(t) := ([nt] + 1 - nt)\frac{S_{[nt]}}{\sqrt{n}} + (nt - [nt])\frac{S_{[nt]+1}}{\sqrt{n}}$$

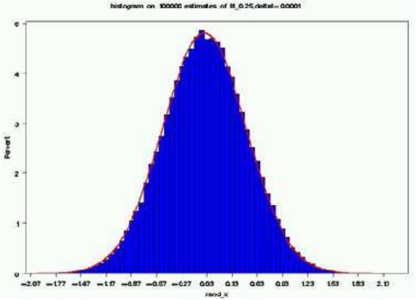
for any $t \in [0, 1]$, which is essentially a linear interpolation (插值) between $[nt]$ and $[nt] + 1$.

- Conclusion: $W^{(n)}(t) \xrightarrow{d} W(t)$.
- Intuition: the central limit theorem (and its abuse?)

Approximation Brownian Motion

The two figures below approximate the distribution of Brownian motion by random walks with $n = 10^3$ and 10^5, respectively.





Geometric Brownian Motion

- For constants (常数) α and σ, define

$$X_t := \alpha t + \sigma W_t,$$

then $\{X_t\}$ is called a Brownian motion with drift.

- For constants α和σ, define

$$X_t := X_0 e^{\alpha t + \sigma W_t},$$

then $\{X_t\}$ is called a geometric Brownian motion (几何布朗运动), whose marginal distribution X_t is log-normal (对数正态分布).

Property 1. Non-differentiable

- Non-differentiable everywhere： Except for a null set (零测集), all other sample paths of W_t are non-differentiable (不可导) at every point in $[0, +\infty)$. (We can also exclude that null set in prior in the definition.)

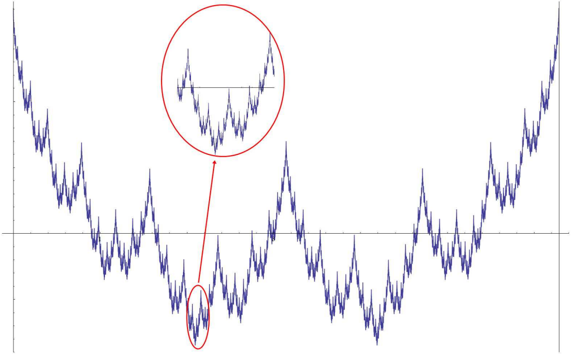
- Note that all the sample paths are continuous...

- Such "weird" function was noticed by K.Weierstrass in 1800s:

$$f(x) = \sum_{n=0}^{+\infty} a^n \cos(b^n \pi x),$$

where we can take 其中$a = 0.5$ and $b = 13$.

Weierstrass's Function



Wiener finds that almost all the continuous functions are "weird" (non-differentiable everywhere).

Property 2. Unbounded Variation

- Unbounded variation (无界变差)：Except for a null set, all other sample paths of W_t are not of bounded variation. More specifically, for any interval $[T_1, T_2]$, we have

$$\lim_{||\Pi|| \to 0} \sum_{j=1}^{n} |W_{t_j} - W_{t_{j-1}}| = +\infty,$$

where $\Pi := \{t_0, t_1, \cdots, t_n\}$ indicates a partition of the interval $[T_1, T_2]$, $t_0 = T_1$, $t_n = T_2$, and

$$||\Pi|| := \max_{1 \le j \le n} |t_j - t_{j-1}|.$$

- Imagination: Brownian motion is the limit of random walk (随机游走).

Property 3. Bounded Quadratic Variation

- Property 2 prevents us from defining the Lebesgue-Stieltjes integral*

$$\int_{T_1}^{T_2} f(t)dW_t.$$

- To overcome this shortcoming, scholars introduced the following property of bounded quadratic variation (平方有界变差) during WWII:

$$\lim_{\|\Pi\|\to 0} \sum_{j=1}^{n} |W_{t_j} - W_{t_{j-1}}|^2 = T_2 - T_1, \quad a.s.$$

Quadratic Variation Process

- Definition. Given a stochastic process $\{X_t\}$, its quadratic variation process (平方变差过程) is defined as

$$[X,X]_t := \lim_{\|\Pi\|\to 0} \sum_{j=1}^{n} |X_{t_j} - X_{t_{j-1}}|^2$$

where $t_0 = 0$ and $t_n = t$.

- Example. Brownian motion: $[W,W]_t = t$.
- In a nonstandard manner, we write $dW_t dW_t = dt$.

Lebesgue-Stieltjes integral*

- Lebesgue-Stieltjes integral in Real Analysis:

$$\int_0^T f(t)dg(t),$$

  which calls for good property of the function $g(t)$.

- If $g(t)$ is monotone on $[0, T]$, then we can prove its differentiability except for a null set, and the define

$$\int_0^T f(t)dg(t) := \int_0^T f(t)g'(t)dt.$$

- A function of bounded variation can be expressed as the difference of two monotone functions.

History of Integrations

- "Integration 1.0" (Riemann, 1854): $\int_0^T f(t)dt$
  - $f(t)$ is Riemann integrable (e.g., continuous).
- "Integration 2.0" (Stieltjes, 1894): $\int_0^T f(t)dg(t)$
  - $g(t)$ is of bounded variation, and $f(t)$ is Lebesgue integrable.
- "Integration 3.0" (Itô, 1944): $\int_0^T X(t)dW(t)$ (??)
  - Sample paths of $W(t)$ are not of bounded variation.

Stochastic Integral of Simple Function

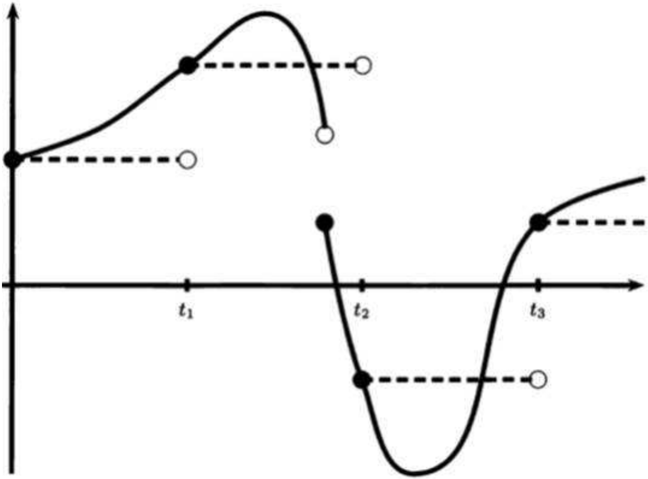
- We have to generalize the definition of integrations. This is a creative task. We start with a simple case – the integrand (被积函数) is a "simple function".
- Simple function has the following form:

$$X(t) = \sum_{j=1}^{n} c_j 1_{[t_{j-1}, t_j)}(t).$$

- The stochastic integral can be defined naturally as:

$$I_X(T) = \int_0^T X(t) dW_t := \sum_{j=1}^{n} c_j (W_{t_j} - W_{t_{j-1}}).$$

Approximation of General Function



Stochastic Integral of General Function

- General function can be approximated by a sequence (序列) simple functions, i.e.,

$$X(t) = \lim_{n\to+\infty} X_n(t).$$

- Stochastic integral: for any $\omega \in \Omega$, define

$$I_X(T)(\omega) := \lim_{n\to+\infty} I_{X_n}(T)(\omega)$$

if the limit on the right-hand side (RHS, 右边) exists.

Notation Map

```
(Simple predictable process)        (Square integrable predictable process)

        C^{(n)}     ----------------------------->  C
    (Cauchy sequence)
           |                                        |
           |                                        |
           |                                        |
           |                                        |
           V                                        V

     I_t(C^{(n)})   -----------------------------> I_t(C)

      (Cauchy sequence)                        (limit exists)
   (by Isometry property)                   (denoted as I_t(C))
                                      (defined as Ito's integral of C)
```

Itô Integral

- Definition of Riemann integral

$$\int_{T_1}^{T_2} f(t)dt := \lim_{\|\Pi\|\to 0} \sum_{j=1}^{n} f(s_j)(t_j - t_{j-1}),$$

where s_j can take any value in $[t_{j-1}, t_j]$.

- Itô integral of an L^2 process $\{X_t\}$ is given by

$$I_X(T) = \int_0^T X_t dW_t := \lim_{\|\Pi\|\to 0} \sum_{j=1}^{n} X_{s_j}(W_{t_j} - W_{t_{j-1}}),$$

where $s_j = t_{j-1}$. The limit on the RHS exists.

An Example

Quiz 9. Compute $\int_0^T W_t dW_t$.

Properties of Itô Integral

- Martingale (defined later)
- linear and continuous operator
- Itô isometry:

$$E[I_X^2(t)] = E\int_0^t X_s^2 ds.$$

- Quadratic variation process:

$$[I_X, I_X]_t = \int_0^t X_s^2 ds.$$

- Theorem*: if the integrand X_t is deterministic, then Itô Integral is a Gaussian process.

Differential Form

- For an Itô integral

$$Y_t := \int_0^t X_s dW_s,$$

conventionally we can re-write it as the following "differential form":

$$dY_t = X_t dW_t.$$

- Differently with canonic calculus, the differential form of Itô integral does not have its own meaning.
- Therefore, the concept "stochastic differentiation" is only a form of abbreviation.

Stochastic Differential Equation

- A stochastic process X_t satisfying

$$X_t = X_0 + \int_0^t \mu(X_s, s)ds + \int_0^t \sigma(X_s, s)dW_s,$$

is defined as diffusion process (扩散过程).

- Differential form:

$$dX_t = \mu(X_t, t)dt + \sigma(X_t, t)dW_t. \quad (1)$$

with an initial value X_0. This is a stochastic differential equation (SDE, 随机微分方程).

- Recall: differential form is only an abbreviation.

The Adaptability of SDE*

- Theorem: in the SDE(1), if there exist such two constants (常数) C and D that

$$|\mu(x,t)| + |\sigma(x,t)| \leq C(1+|x|),$$

$$|\mu(x,t) - \mu(y,t)| + |\sigma(x,t) - \sigma(y,t)| \leq D(|x-y|),$$

  then SDE(1) has a unique strong solution.

- Example: the Black-Scholes model; CEV model when $\beta = -1$.

Itô's Lemma

- Principle: for a smooth function f, we have:

$$df(X_t) = f'(X_t)dX_t + \frac{1}{2}f''(X_t)d[X,X]_t.$$

- Itô's Lemma: for a bivariate smooth function f, the process $f(W_t,t)$ satisfies:

$$df(W_t,t) = f_1(W_t,t)dW_t + \left[f_2(W_t,t) + \frac{1}{2}f_{11}(W_t,t)\right]dt.$$

- Replace the chain rule (锁链公式) in canonic calculus. Red terms are all caused by the non-zero quadratic variations. Extension:

$$df(X_t,t) = f_1(X_t,t)dX_t + f_2(X_t,t)dt + \frac{1}{2}f_{11}(X_t,t)d[X,X]_t.$$

The Black-Scholes model

- Itô's Lemma enables one to verify immediately that the geometric Brownian motion (几何布朗运动)

$$S_t := S_0 e^{\alpha t + \sigma W_t}$$

is governed by the following SDE

$$\begin{cases} dS_t = \mu S_t dt + \sigma S_t dW_t, \\ S_0 = s_0. \end{cases}$$

This is known as the Black-Scholes Model (BSM), summarized by P. Samuelson and other scholars around 1964.

Why BSM

- Empirical analysis at that time showed that the relative returns of stocks are basically stable, and their correlations among different periods are weak.

$$\frac{S_{t+\Delta t} - S_t}{S_t} \approx \mu \Delta t + \sigma (W_{t+\Delta t} - W_t).$$

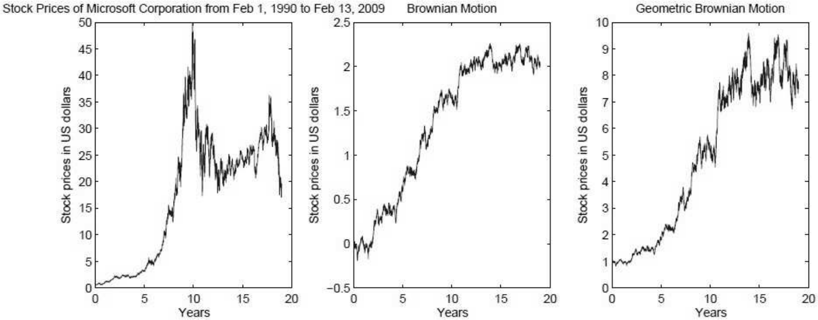
- Relative returns:

$$E\left[\frac{S_{t+\Delta t} - S_t}{S_t}\right] = \mu \Delta t.$$

- Variance:

$$Var\left(\frac{S_{t+\Delta t} - S_t}{S_t}\right) = \sigma^2 \Delta t.$$

Fluctuations of Stock Prices



Diffusion Processes

- Diffusion process is an important category of asset price models.
    - The BSM: $dS_t = \mu S_t dt + \sigma S_t dW_t$
    - CEV model: $dS_t = \mu S_t dt + \sigma S_t^{1+\beta} dW_t$
    - CIR model: $dr_t = (\alpha - \beta r_t) dt + \sigma \sqrt{r_t} dW_t$
    - Stochastic volatility models (two dimensional SDE)
    - Jump factor
- New models improve the shortcomings of the BSM. See Chapter 3.

Solve An SDE

- Similarly with deterministic differential equations (ODEs and PDEs), most SDE do not possess any explicit solution, which forces the practitioner to adopt numerical procedures; see Chapter 4.
- Only a few SDEs have closed-form solutions. Example: the BSM.

$$d \ln S_t = \frac{1}{S_t} dS_t + \frac{-1}{2S_t^2} d[S,S]_t = \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma dW_t.$$

- Recall: the quadratic variations of Brownian motion and Itô integral.

Exercises

- Quiz 10. Write the SDE of geometric Brownian motion S_t.

- Quiz 11. Compute dW_t^n, where $n \neq 0, 1$.

Why to Study Option Pricing

- "Ancient options" were not active, and thus had low liquidity. The quotes are highly random.
- CBOE founded in 1973 initiated standard option contracts, which started a blooming financial industry. European options are simple and liquid, and provide significant information in the market. Hence, researchers focused on European options first.
- Theoretically, wrong prices lead to arbitrage, which, in turn, correct the prices.
- Replication (复制): strategy of risk management

Comparison: Forward and Option

- Difference between the value and the strike
- Option pricing: for fixed S_t, t, T, and K, determine the value of an option contract.
- Forward pricing: for fixed S_t, t, T, and the conventional initial value 0 of the forward contract, determine the appropriate strike K.
- Forward pricing is irrelevant with the information of the stochastic process $\{S_t\}$, and is thus completely objective. On the contrary, option pricing depends heavily on the model of $\{S_t\}$, and is thus subjective.

Intrinsic Value and Time Value

- Recall: the terminal payoff of a financial contract is the revenue of its holder at the maturity.
- Examples: call option $(S_T - K)^+$; put option $(K - S_T)^+$.
- The option price can be decomposed into two parts: intrinsic value (内在价值) and time value. Intrinsic value is defined as the payoff if we pretend that we can exercise this contract right now.

$$P(S_t, t, K) = (K - S_t)^+ + \left[P(S_t, t, K) - (K - S_t)^+ \right]$$

In-the-money and Out-of-the-money

- A call option is called as in-the-money (实值、价内) if $S_t > K$, and out-of-the-money (虚值、价外) if $S_t < K$.
- When $S_t \approx K$, the option is at-the-money (平值).
- Deep in-the-money/out-of-the-money: when S_t and K are far away.
- Reversed for put options

Assumption: Frictionless Market

Option pricing theory assumes the market is frictionless:

- no trading cost
- no tax
- many traders and no one can manipulate the market
- equality on entering the market
- no information cost
- common market expectation
- no crisis cost
- no bid-ask spread
- no interest rate spread

Basic Idea of Option Pricing

- Assumption on the evolution of asset price (model)
- Model parameter estimation based on market data
- Option price must be the expectation of its terminal payoff, under "some probability"
  - risk-neutral expectation
- If this expectation cannot be expressed explicitly, one shall adopt numerical algorithm to approximate the true option price.

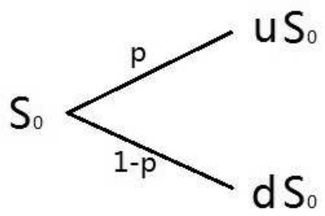
Fair Gambling Rule

- A gambling (赌博) with a random profit X is called a fair gambling if $E[X] = 0$.
- For example: toss a coin:
  - head, then you win \$1;
  - tail: then you win \$0.

  This game is a fair gambling iff the ticket is \$0.5.
- The fair gambling principle (公平赌博原则): a financial contract V with a random terminal payoff X on the maturity T must (???) have the initial price

$$V_t = e^{-r(T-t)} E\left[V_T\right].$$

A Toy Model: Single-Period Binary Tree



- Underlying asset S has two possible prices at time 1.
- A derivative V of S has two possible payoffs V_u and V_d at time 1, determined by the price of S.
- Fair gambling: $V_0 = e^{-r}[pV_u + (1-p)V_d]$. (???)

Arbitrage Pricing

- Construct a portfolio $\Pi := \alpha S - V$, whose two possible values at time 1 coincide.
- Solve $\Pi_{1u} = \Pi_{1d}$ and obtain:

$$\alpha u S_0 - V_u = \alpha d S_0 - V_d, \quad \Rightarrow \quad \alpha = \frac{V_u - V_d}{u - d}\frac{1}{S_0}.$$

$$\Pi_{1u} = \Pi_{1d} = \frac{dV_u - uV_d}{u - d} = e^r \Pi_0.$$

- Arbitrage Pricing:

$$V_0 = \alpha S_0 - \Pi_0 = \frac{(e^r - d)V_u + (u - e^r)V_d}{e^r(u - d)}.$$

The Risk-Neutral Measure

- Note that the result holds under any measure (any value of p).
- Define $q := (e^r - d)/(u - d)$, then we have
$$V_0 = e^{-r}[qV_u + (1-q)V_d],$$
which is essentially a new probability measure Q.
- Under Q, all of the asset price can be expressed as the discounted expected payoff:
$$V_t = e^{-r(T-t)}E_Q[V_T].$$
Such measure Q is known as the risk-neutral measure (风险中性测度).

From Binary Tree to Continuous Model

- Although the single-period binary tree model is quite simple, it holds the essential features of complicated continuous models:
  - random payoff;
  - fair gambling;
  - risk-neutral measure.
- Under continuous-time dynamic models, we still aim on the risk-neutral measure to finish the option pricing. The change of measure will not be operated on two branches, but on all the sample paths of Brownian motion.

Martingale*

- For $\mathscr{T} = [0, +\infty)$, a stochastic process $\{X_t : t \geq 0\}$ is defined as a martingale (鞅), if:
  - 1. for any $t \geq 0$, we have $X_t \in L^1(\Omega)$;
  - 2. $\{X_t\}$ is an "adapted" process on the "filtration" $\{\mathcal{F}_t\}$;
  - 3. for any $0 \leq s \leq t$, we have

$$E[X_t|\mathcal{F}_s] = X_s.$$

- Examples of martingales:
  - Brownian motion $\{W_t\}$;
  - geometric Brownian motion $\{e^{\sigma W_t - \frac{1}{2}\sigma^2 t}\}$.

Girsanov Theorem*

W_t is a standard Brownian motion under measure P. For some stochastic process $\Theta(t)$, define

$$\tilde{W}(t) := W(t) - \int_0^t \Theta(s)ds$$

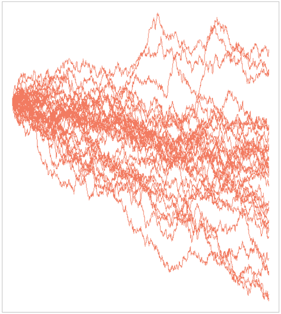
which is a Brownian motion with drift, and

$$Z(t) := \exp\left\{\int_0^t \Theta(s)dW_s - \frac{1}{2}\int_0^t \Theta^2(s)ds\right\},$$

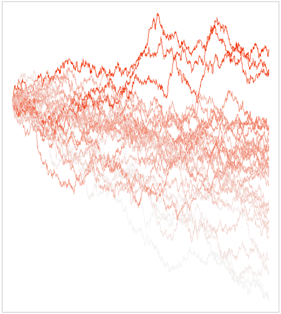
$$\left.\frac{dQ}{dP}\right|_{\mathscr{F}_t} = Z(t).$$

Then $\tilde{W}(t)$ is a standard Brownian motion under the new measure Q.

Change of Measure



30 paths of a Brownian motion with negative drift



The same paths reweighted according to the Girsanov formula

Change of Measure for the BSM

- The BSM:

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

becomes

$$dS_t = rS_t dt + \sigma S_t d\tilde{W}_t$$

under the new measure Q.

- The process $\Theta(t)$ in Girsanov Theorem takes a constant

$$\Theta(t) \equiv -\frac{\mu - r}{\sigma}.$$

- $\Theta(t)$ is the market price of risk (风险市价), indicating the price of purchasing one unit of the risk.

Black-Scholes Option Pricing Formula

- Under Q, we solve the BSM and obtain

$$S_T = S_0 \exp\left\{\left(r - \frac{\sigma^2}{2}\right)T + \sigma \tilde{W}_T\right\}.$$

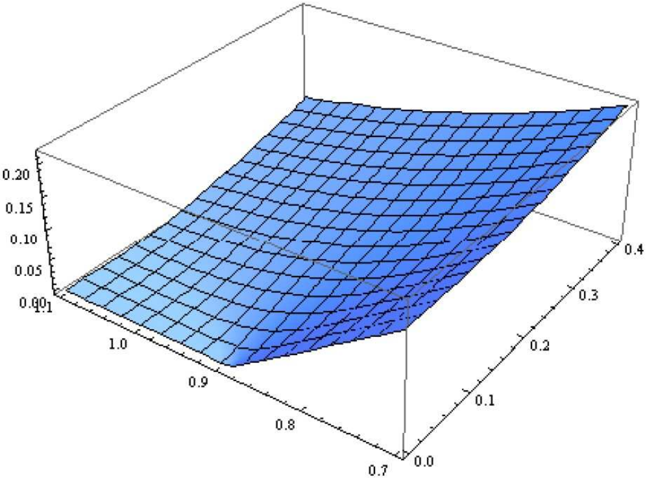
- Risk-neutral pricing of European call options:

$$\begin{aligned} V_0 &= e^{-rT} E_Q[V_T] \\ &= e^{-rT} E_Q[(S_0 e^{(r-\frac{\sigma^2}{2})T+\sigma\tilde{W}_T} - K)^+] \\ &= S_0 \cdot N(d_1) - e^{-rT} K \cdot N(d_2), \end{aligned}$$

where $d_1 = \frac{\ln(S_0/K)+(r+\sigma^2/2)T}{\sigma\sqrt{T}}$, $d_2 = \frac{\ln(S_0/K)+(r-\sigma^2/2)T}{\sigma\sqrt{T}}$.

- Note that μ disappears, just like p in the single-period binary tree model...

A Figure of Option Prices



Feynmann-Kac Theorem*

- The option pricing formula can be also derived by self-financing portfolio and arbitrage pricing, which depends heavily on stochastic calculus.

- Feynmann-Kac Theorem. The underlying asset process $\{S_t\}$ follows a diffusion process under Q, and its terminal payoff has a form $V(T,S)=G(S)$. Then the bivariate function $V(t,x)$ satisfies the following parabolic PDE of the second order (二阶抛物型偏微分方程):

$$V_t(t,x)+\mu(t,x)V_x(t,x)+\frac{1}{2}\sigma^2(t,x)V_{xx}(t,x)=rV(t,x),$$

with $V(T,x)=G(x)$.

Summary: Principle of Martingale Pricing

- Under the risk-neutral measure Q, discounted price process of any financial asset $e^{-rt}V_t$ is a martingale.
- Risk-neutral pricing:

$$V_t = e^{-r(T-t)} E_Q[V_T | \mathcal{F}_t].$$

- The change of measure does not change any sample path, and only changes the subjective probability weights.
- Valid for non-BSM models, non-European options and other assets.

An Elegant Example*

- This is an example of exotic option pricing under general models. Assume the underlying asset pricing process $\{S_t\}$ follows the diffusion model

$$dS_t = (r - q)S_t dt + \sigma(S_t)S_t d\tilde{W}_t$$

under risk-neutral measure Q, where $\sigma(x)$ is a deterministic function. A derivative contract V pays 1 when S_t up-crosses the level $U(> S_0)$ for the first time.

- Risk-neutral pricing:

$$V_0 = E_Q[e^{-r\tau_U} 1_{\{\tau_U < \tau_0\}}],$$

where τ_U and τ_0 are first passage times (首达时) of the process $\{S_t\}$ to the levels U and 0, respectively.

An Elegant Example (Cont'd)*

The following result is proposed by Davydov and Linetsky (2001, MS). Let $\psi(S)$ be one of the basic solutions to the ODE

$$\frac{1}{2}\sigma^2(S)S^2\frac{d^2f}{dS^2}+\mu S\frac{df}{dS}-rf=0,$$

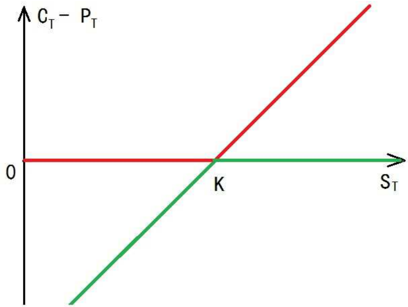
then the price of the derivative V is given by

$$V_0=\frac{\psi(S_0)}{\psi(U)}.$$

Particularly, if $\sigma(x)$ is a power function, i.e., $\{S_t\}$ follows a CEV model, then the function $\psi(\cdot)$ has closed-form.

The Put-Call Parity

Consider a portfolio: long a call option and short a put option. Its terminal payoff is shown as the figure below.



The put-call parity:

$$C(S_t, t, K) - P(S_t, t, K) = S_t - Ke^{-r(T-t)}.$$

Stopping Time

- On a filtered* probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\in\mathscr{T}}, P)$, an $\mathcal{F}$ measurable random variable T is called as a stopping time (停时), if

$$\{T \leq t\} \in \mathcal{F}_t$$

for all $t \in \mathscr{T}$. Here the value range of T is $[0, +\infty]$.

- Examples: a constant; first passage time into a closed set.

- The meaning of a stopping time: when the random time arrives, we are informed that it arrives.

Optimal Stopping Theory*

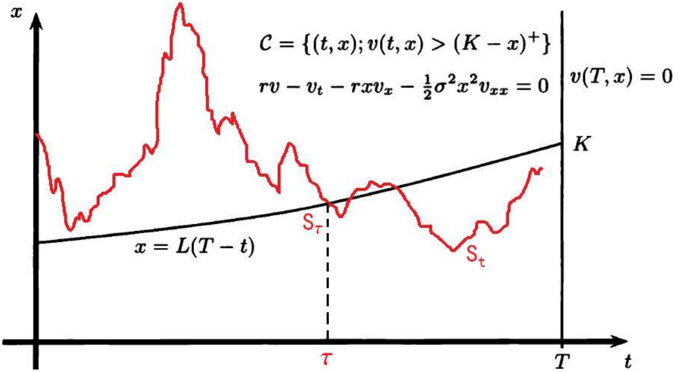
- In game theory (博弈论), the optimal stopping is a strategy, designed for recruitment, blind dating, estate selling, nuclear war triggering, etc.
- In mathematics, the optimal stopping theory indicates the theory that solves a stopping time to optimize a stochastic system.
- Example: the exercise strategy of an American option should be a stopping time, which can thus be solved via the optimal stopping theory.
- Analogy. You meet many people in your life. Some of them are "able" to fall in love with you. Timing (时机) is often crucial to determine who you get.

American options

- American options are also standard contracts traded in markets. Differently with European option, American option allows its holder to exercise the option at or before the maturity T.
- Reminder: exercise (执行) is different with closing positions (平仓).
- Discrete form: Bermudan option allows its holder to exercise the option at several time points.
- Theorem. American call option and European call option (with the same parameters S_0, K, t and T) have the same price, if there is no dividend (股息).
    - Proof: arbitrage portfolio.

Exercise of American Put Option

American put option (and call option when dividend exists) is difficult to price. It is described by the optimal stopping theory, which provides an "optimal exercise boundary".



Exotic Options

- OTC options are also known as exotic options (奇异期权). Some of them are path-dependent options, whose payoff depends the sample path of $\{S_t\}$.

- Lookback options; Asian options; barrier options...

- The payoff of lookback option depends on the running maximum or minimum of the underlying process $\{S_t\}$:

$$V(T) = \left(\inf_{t \in [0,T]} S_t - K \right)^+ ; \quad V(T) = \left(K - \sup_{t \in [0,T]} S_t \right)^+ .$$

Asian Options

- The payoff of Asian options depends on the average value of the sample path of $\{S_t\}$:

$$V(T) = \left(\frac{1}{T}\int_0^T S_t dt - K\right)^+$$

- Advantages:
  - stable price, difficult to manipulate
  - convenient risk management for those companies with periodic revenue
  - lower price than corresponding European option
- Difficult to price even under the BSM. Geman and Yor (1993, MF) proposes a PDE approach with 3-fold numerical integrals.

Discretely Monitored Asian Options

- Discretely monitored Asian option calculates the arithmetic average of the underlying asset prices at several time points (e.g., the end of every week):

$$V(T) = \left(\frac{1}{m+1} \sum_{k=0}^{m} S_{k\Delta} - K \right)^{+}$$

- It is even more difficult to price, recently approximated via frontier theories such as Malliavin Calculus. See, e.g., Cai, Li and Shi (2014, MOR)

- Curse of dimensionality (维数灾难)

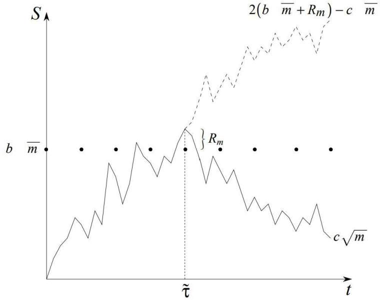
Barrier Options

- The payoff of barrier option is similar with European option, except that a barrier option becomes valid (knock-in) or invalidated (knock-out) immediately when the process $\{S_t\}$ up-crosses or down-crosses a pre-specified barrier level.
- 8 types: call and put; upstream and downstream barrier; knock-in and knock-out. Example: down-and-out put options.
- Cai, Li and Shi (2021, JEDC) studies the pricing algorithm of discretely monitored barrier options:

$$V(T) = (K - S(T))^{+} 1_{\{S(t_1)>B,\, S(t_2)>B, \cdots, S(t_n)>B\}}$$

Discretely Monitored Barrier Options

Famous progress：Broadie, Glasserman, and Kou (1997, MF) introduces the Riemamm Zeta function in the number theory (数论), to approximate the price of discretely monitored barrier option by usual barrier options under the BSM, in a very smart and graceful manner.



Greeks

- Greeks (希腊值) are some partial differentiations (偏导数) of derivative prices, also known as sensitivities.
- The most popular Greek is Delta, the partial differentiation with respect to S_t:

$$\Delta := \frac{\partial V(S_t, K, \mu, \sigma, t, T, r)}{\partial S_t}.$$

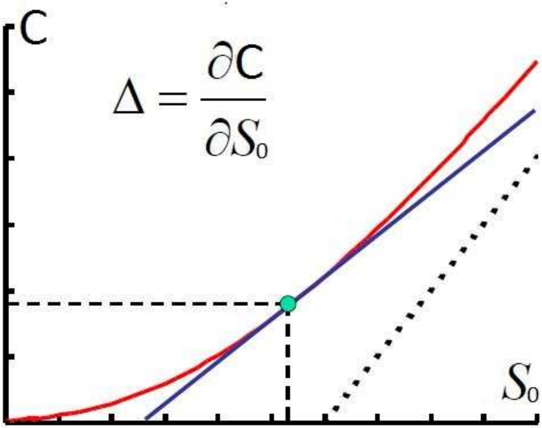
- Other Greeks:

$$\Gamma := \frac{\partial^2 V}{\partial S_t^2}, \quad \Theta := \frac{\partial V}{\partial t}, \quad \text{(Vega)}\ \mathcal{V} := \frac{\partial V}{\partial \sigma}, \quad \rho := \frac{\partial V}{\partial r}.$$

Delta Hedging

- Risk management, the perfect hedging ratio
- Decrease, or even eliminate the uncertainty.
- Example: a portfolio $C_t - \Delta_t S_t$ replicates a riskless bond.
- In practice, there exists hedging errors due to the limitation on trading frequency and computational errors.

Delta of European Call Options



Values of Delta

- Delta of European call option under the BSM: $\Delta = N(d_1)$, increases from 0 to 1.
- Delta of European put option under the BSM: $\Delta = -N(-d_1)$, increases from -1 to 0.
- Differ among different models.

Signs of Greeks for European Options

Greeks	Call	Put
Δ	+	-
Γ	+	+
Θ	-	ITM + ; OTM -
$\mathcal{V}$	+	+
ρ	+	-